



Photosynthetic induction and sunfleck responses of three understory tree species in forests of the Mid-Atlantic U.S

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Abstract

Key message Photosynthetic induction was rapid in three shade-tolerant understory saplings, and surprisingly, native and non-native *Acer* species gained induction more quickly when exposed to sequential lightflecks, relative to constant saturating light.

Abstract Growing regeneration mismatch between forest canopy and understory environments is evident in broadleaf forests of the eastern U.S., leading to novel understory forest composition. Three co-occurring shade-tolerant tree species in forest understories of southeastern Pennsylvania are the natives *Acer rubrum* and *Fagus grandifolia* and the non-native *Acer platanoides*. Physiological attributes contributing to these species' relative success in the understory's dynamic light environment are poorly characterized. Three saplings per species from three sites were sampled to examine photosynthetic induction and lightfleck responses. Predictors of photosynthesis (A) and stomatal conductance (g_s) during lightflecks were also examined. The rate and magnitude of photosynthetic induction were similar among species, with all gaining induction rapidly, as expected for shade-tolerant species. However, physiological strategies supporting induction responses varied by species, as *A. rubrum* had more responsive stomata and significantly lower stomatal limitation during induction, as well as significantly higher A/g_s when fully induced. Surprisingly, lightfleck responses differed from induction responses for the two *Acer* species, both of which exhibited significantly greater cumulative carbon assimilation than when exposed to constant saturating light, particularly during short-duration lightflecks. Leaf nitrogen and g_s were significant predictors of lightfleck responses, while species identity was not. Although induction gain under saturating light was rapid for all species, variable lightfleck responses favoring the *Acer* species provide an advantage given the intermittent nature of understory light availability. As eastern broadleaf forest composition continues to change in response to diverse pressures, the *Acer* species are well-positioned to persist.

Keywords Broadleaf forest · Forest understory · Leaf traits · Lightfleck · Photosynthetic induction · Stomatal conductance

Introduction

Broadleaf forests of the eastern U.S. are undergoing substantial change as pressures including mesophication, changing climate, fragmentation, white-tailed deer (*Odocoileus virginianus*) overabundance, and invasive species act on these ecosystems (Nowacki and Abrams 2008; McEwan et al. 2011; Miller and McGill 2018). One well-documented outcome of

these pressures is a growing mismatch between forest canopy and understory tree species composition (McEwan et al. 2011; Miller and McGill 2018; Vickers et al. 2019). This mismatch is particularly severe in the U.S.'s Mid-Atlantic region in forests historically dominated by oaks (*Quercus* sp.) and hickories (*Carya* sp., Miller and McGill 2018, Vickers et al. 2019). Tree species now occupying the understory are typically better adapted to mesic conditions (often *Acer rubrum*, *Acer saccharum*, McEwan et al. 2011) than those in the canopy and include species resistant to deer browse (e.g. *Fagus grandifolia*, *Prunus serotina*, Miller and McGill 2018; Webster et al. 2018). Novel understory conditions, in which there is regeneration mismatch, as well as regeneration failure, are well documented in the U.S.'s Mid-Atlantic region (Miller and McGill 2018), and as these conditions emerge,

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many questions remain regarding the long-term trajectory of these understory tree communities. Better understanding the physiological attributes of trees inhabiting the understory is important, as the success or failure of these trees will significantly impact forest development over the next century.

Trees in the forest understory encounter a dynamic light environment, in which diffuse light is punctuated by what are often short-duration sunflecks that can provide 45–80% of daily understory photosynthetic photon flux density in temperate broadleaf forests (Chazdon 1988; Miyashita et al. 2012; Way and Pearcy 2012). Such conditions create photosynthetic challenges as factors including carbon gain, water loss, and energy expenditure must be balanced (Percy 1990; Vico et al. 2011). The process of photosynthetic induction is an important component of a leaf's ability to respond to a rapidly changing light environment. Both biochemical and stomatal processes regulate induction responses, with light-induced enzyme activation, the buildup of Calvin Cycle intermediates, and stomatal responsiveness all playing roles (Way and Pearcy 2012). Variation in these processes can affect a leaf's ability to respond readily to sunflecks, and woody plants exhibit a variety of photosynthetic induction responses that do not always coincide with hypothesized responses for a given species (discussed below, Naumburg and Ellsworth 2000; Urban et al. 2007; Vico et al. 2011; Way and Pearcy 2012). As novel understory tree communities develop throughout the eastern U.S. and species begin to compete in light-limited environments, sunfleck responses and the ability of leaves to become quickly induced are important components in assessing the likelihood of a species' overall success. Species that quickly gain induction and respond readily to temporally patchy sunflecks will likely experience enhanced carbon gain (Naumburg et al. 2001), promoting long-term growth and survival in these light limited environments.

Many efforts have been made to characterize photosynthetic induction and sunfleck responses based on a tree's life history characteristics and growth environment, but these studies have yielded mixed results (Percy and Way 2012; Way and Pearcy 2012). Of particular relevance to the present study, shade tolerant species are expected to rapidly gain induction in response to sunfleck exposure and maintain induction following a return to low-light conditions, relative to shade intolerant species (Naumburg and Ellsworth 2000; Vico et al. 2011). This hypothesis establishes an expectation that shade-tolerant saplings in forest understory environments will have similar induction and sunfleck responses. However, such an assumption is not consistently borne out in the literature (Naumburg and Ellsworth 2000; Urban et al. 2007; Vico et al. 2011; Way and Pearcy 2012), suggesting the need to explore variation in these characteristics among species, as well as the leaf traits underlying this variation (Wachendorf and Küppers 2017). Such an approach may

provide a more robust understanding of factors governing variation in photosynthetic induction and sunfleck responses among shade tolerant trees.

The present study took place in the fragmented forest ecosystems of southeastern Pennsylvania, an area with both high regeneration mismatch and failure (Miller and McGill 2018). Only 29% of the region's land area remains in forest cover, with 32% of that area comprised of fragments ≤ 1 ha (Albright et al. 2017). Throughout Pennsylvania, *A. rubrum* and *F. grandifolia* are abundant in seedling, sapling, and small diameter (< 10 cm dbh) tree size classes (Albright et al. 2017), while regionally, *Acer platanoides* is the most common non-native tree species in small forest fragments (Schedlbauer and Polohovich 2020; Trammell et al. 2020). The occurrence of *A. platanoides* on forest land throughout the northern U.S. has increased in recent years (Kurtz 2023), and the presence of this species may exacerbate regeneration mismatch. *Acer platanoides* is known to suppress the growth of native tree species (Galbraith-Kent and Handel 2008; Martin et al. 2010; Fang and Wang 2011, 2020), so it will likely persist in the forest fragments where it becomes established.

Given the novel forest understory taking shape in southeastern Pennsylvania, co-occurring *A. platanoides*, *A. rubrum*, and *F. grandifolia* saplings were selected for examination of their photosynthetic induction and sunfleck responses. All three species are shade-tolerant (Niinemets and Valladares 2006), though *F. grandifolia* saplings exhibit higher survivorship in deep shade, followed by *A. platanoides*, then *A. rubrum* (Kobe et al. 1995; Martin et al. 2010). *Acer platanoides* has well documented invasive traits (after van Kleunen et al. 2010), including high height and diameter growth under moderately low to high light, relative to both native species (Martin et al. 2010). *Acer platanoides*' presence in either the forest canopy or sapling community can suppress height growth of native saplings, including *A. rubrum*, while conspecific saplings remain unaffected (Galbraith-Kent and Handel 2008). These findings suggest that *A. platanoides* may possess physiological attributes, including photosynthetic induction and sunfleck responses, that differ markedly from native species, including the non-native's congener *A. rubrum*.

To explore leaf-level physiological differences among these three tree species within forest understory environments, the following research questions were addressed: (1) Do photosynthetic induction responses vary among species? (2) Do biochemical vs. stomatal limitations on the induction process vary by species? (3) Do the study species vary in their responses (both photosynthesis and stomatal conductance) to artificially applied sunflecks, and does sunfleck duration affect these responses differentially? (4) What are the key predictors of photosynthesis during artificially applied sunflecks? Exploration of these questions will

improve our understanding of the physiological attributes contributing to these species' success in forest understory environments, while also elucidating controls on induction and sunfleck responses in shade-tolerant tree saplings.

Methods

Study sites and tree characteristics

Four temperate broadleaf forests sites in southeastern Pennsylvania were selected for use in this study. Sites are within 20 km of one another and included the Gordon Natural Area (GNA, 39° 56' 08" N, 75° 35' 55" W), Miller Farm (MF, 40° 01' 54" N, 75° 43' 29" W), Mount Bradford Preserve (MBP, 39° 56' 33" N, 75° 37' 26" W), and Binky Lee Preserve (BLP, 40° 05' 42" N, 75° 36' 00" W). Forest canopies at these sites are dominated principally by oak and hickory species, as well as tulip poplar (*Liriodendron tulipifera*). Soils at the sites are Gladstone gravelly loams (GNA, MF, BLP) and Parker gravelly loams (MBP, NRCS 2022). The region's annual temperature averages 11.5 °C, with a mean monthly minimum of − 1.4 °C in January and a mean monthly maximum of 23.8 °C in July (NOAA 2012). Total annual precipitation averages 1237 mm and is approximately evenly distributed throughout the year (NOAA 2012).

Three understory tree species were investigated in the present study: Norway maple (*A. platanoides*), red maple (*A. rubrum*), and American beech (*F. grandifolia*). Three trees per species were measured at each of three closed-canopy study sites, yielding measurements of nine trees per species. However, not all sites contained three understory individuals of all three species with similar size and physical proximity to one another. An additional requirement was that all measurement trees have attached foliage that could be measured from the ground or via a ladder. Therefore, measured trees (always three individuals per species at a site) were spread across the four study sites. *Acer platanoides* was measured at BLP, GNA, and MPB, *A. rubrum* was measured at BLP, GNA, and MF, and *F. grandifolia* was measured at GNA, MBP, and MF. Measured trees averaged 6.4 cm diameter at breast height (range: 4.1–9.7 cm), while height averaged 6.3 m (range: 3.4–11.4 m).

Field measurements

Hemispherical photographs

The understory light environment of each measured tree was characterized via hemispherical photography in July 2020. Photographs were taken on uniformly overcast days with a Nikon D40 DSLR camera (Nikon Corporation, Tokyo, Japan) and circular fisheye lens (4.5 mm F2.8 EX DC Sigma,

Kanagawa, Japan). The camera was positioned on the south side of each sample tree, approximately 1 m from the edge of the tree's canopy and 1.5 m above the ground. The camera was leveled at each location, with the top of each image oriented toward true north. All photographs were taken in shutter priority mode with a shutter speed of 1/125 s. Image analysis was performed with HemiView (v. 2.1, Delta-T Devices, Ltd., Cambridge, UK) to determine percent canopy cover and global site factor (GSF), which is the proportion of global radiation (i.e., direct plus diffuse radiation) reaching a given location, relative to a location with no sky obstruction.

Light response curves

In late June 2020, light response curves were measured on the leaves of three trees per species at the GNA site to determine an appropriate setting for saturating light during subsequent measurements. All measured leaves were fully expanded and attached to unique understory trees, growing under shaded conditions. The LI-6800 infrared gas analyzer fitted with the 9 cm² cuvette and red/blue light source (6800-02) was used for this and all other gas exchange measurements (LI-COR, Inc., Lincoln, NE). Light response curves were initiated at high light (1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$), with measurements logged at light intensities of 1000, 800, 600, 300, 150, 50, and 0 $\mu\text{mol m}^{-2} \text{s}^{-1}$. Photosynthesis in all species was light-saturated, with no signs of photoinhibition at a light intensity of 1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ (data not shown).

Photosynthetic induction and lightfleck experiments

All gas exchange measurements described hereafter were made on uninduced, fully expanded leaves, positioned ~ 2–3 m above the ground, and maintained in darkness for at least 12 h prior to measurement. Darkness was achieved by securing black nylon bags that block 97% of radiation in direct sunlight (Schedlbauer 2015) over branch tips. Measurements of gas exchange during photosynthetic induction, as well as during simulated sunfleck experiments were made for each sample tree on separate, attached leaves between late June and late July 2020. All measurements were made between 9:00 and 16:00 standard time. Uninduced leaves were clamped into the leaf cuvette and exposed to low light conditions (10 $\mu\text{mol m}^{-2} \text{s}^{-1}$), intended to simulate the shaded forest understory environment. These conditions were maintained until photosynthesis (A) and stomatal conductance to water vapor (g_s) were stable for 5 min, after which one of three different experiments was initiated to measure induction or sunfleck responses. For all measurements, cuvette CO₂ concentration was maintained at 410 $\mu\text{mol mol}^{-1}$, with a relative humidity of 55%. Cuvette temperature was allowed to vary with air temperature

between 25 and 30 °C, leading vapor pressure deficit to vary between 1.43 and 1.91 kPa during measurements.

A photosynthetic induction experiment recorded gas exchange for 2 min at low light (where A varied by $<0.15 \mu\text{mol m}^{-2} \text{s}^{-1}$, with two exceptions up to $0.23 \mu\text{mol m}^{-2} \text{s}^{-1}$, and g_s varied by $<0.0050 \text{mol m}^{-2} \text{s}^{-1}$, with one exception at $0.0070 \text{mol m}^{-2} \text{s}^{-1}$), followed by measurements at saturating light ($1000 \mu\text{mol m}^{-2} \text{s}^{-1}$) until A and g_s were again stable (i.e., variation in $A < 0.15 \mu\text{mol m}^{-2} \text{s}^{-1}$, variation in $g_s < 0.0060 \text{mol m}^{-2} \text{s}^{-1}$) for 5 min (after Naumburg and Ellsworth 2000). Gas exchange measurements were recorded every 2 s for the first 4 min and every 10 s thereafter during induction, and these measurements are referred to as a 'trace' for a given leaf.

The artificial sunfleck (hereafter referred to as lightfleck) experiments also began at low light ($10 \mu\text{mol m}^{-2} \text{s}^{-1}$) with 2 min of measurements followed by a series of five saturating lightflecks ($1000 \mu\text{mol m}^{-2} \text{s}^{-1}$), each separated by 2 min of low light. Two lightfleck experiments were performed; in one, a series of 30 s lightflecks were applied, while in the other 60 s lightflecks were applied (after Chazdon and Pearcy 1986). Gas exchange measurements were recorded every 2 s during all lightfleck experiments. Following completion of measurements, all leaves were collected and transported to the lab on ice.

Laboratory analyses

Seventeen of the 81 gas exchange measurements described above were area-corrected due small necrotic spots ($\sim 1\text{--}2 \text{ mm}$ diameter) on leaf tissue within the cuvette. Living (green) leaf area was determined for scans of the 9 cm^2 of leaf tissue contained within the cuvette using ImageJ (v. 1.53a, NIH, Bethesda, MD). In addition, whole leaf blades for each of the 81 measurements were scanned to determine total leaf area using WinFolia (Regent Instruments, Montréal, Québec) and dried for at least 48 h at 70 °C to determine leaf mass. Leaves were then ground to a fine powder with an oscillating ball mill (MM 200, Retsch, Haan, Germany) and packed into tin capsules for percent nitrogen (N) analysis at the Washington State University Stable Isotope Core Laboratory. Nine duplicate samples analyzed to quantify error had a mean absolute difference of 0.05% N. These data were used to compute dry leaf mass per unit leaf area (LMA, g m^{-2}), as well as nitrogen per unit leaf area (N_{area} , g N m^{-2}).

Of the 81 whole leaves, 37 had necrotic spots on the leaf surface (this number differs from that above because it was often possible to avoid necroses when clamping a leaf into the cuvette). Given their small size, it was unfeasible to account for necroses in LMA determination. Necrotic spots on these 37 leaves accounted for an average of 2.7% of total leaf area, with only 4 leaves covered by $>5\%$ necrotic area.

Necrotic tissue was excluded from the leaf tissue analyzed for percent N.

Data analyses

All analyses described below were performed with the open source software R (v. 4.4.1, R Core Team 2024). Both GSF and percent canopy cover were analyzed with two-way ANOVA where the main effects were site, species, and their interaction. Leaf trait data, specifically LMA and N_{area} determined for leaves compiled from all three experiments, were analyzed separately using linear mixed-effects models with the fixed effect species and random effect study site. Any significant ($\alpha = 0.05$) fixed effect detected via ANOVA was further investigated with a Tukey test. Transformations as indicated by the Box-Cox method (Box and Cox 1964) were required to meet normality assumptions for both LMA ($y^{-0.5}$) and N_{area} ($1/y$).

The process of photosynthetic induction was modeled separately for each species with a logistic regression function,

$$y = Y_{\text{asym}} / (1 + \exp(-k(x - t_m))) \quad (1)$$

where y is photosynthetic rate ($\mu\text{mol m}^{-2} \text{s}^{-1}$), x is time elapsed (s), Y_{asym} is the model's asymptote or maximum value of y , k describes the positive slope of model's curve, and t_m is the time at which the rate of increase in the positive slope is maximized. In addition to separate models by species, one overall model was fit using data from all three species. Individual species models were then compared to the overall model via whole-model F-tests to determine whether any species-level models differed significantly from the overall model. Note that individual leaf traces for this analysis and those described below were subject to post-field data processing, as some traces were logged for longer than necessary (see Online Resource 1).

Individual leaf traces were used to compute a one-point estimate of the maximum rate of Rubisco carboxylation (V_{cmax}), after De Kauwe et al. (2016), and based on terms derived from Collatz et al. (1991), Bernacchi et al. (2001), and Crous et al. (2013). This method assumes A_{max} is limited only by the carboxylation capacity of Rubisco, with no limitation due to ribulose-1,5-bisphosphate (RuBP) regeneration. Given a V_{cmax} estimate for each leaf, it was then possible to determine delays in photosynthetic induction that were due to biochemical or stomatal limitations. The differential limitation analysis method of Deans et al. (2019) was used to determine the proportion of each leaf's induction curve subject to biochemical or stomatal limitations (see Online Resource 1 for more details).

Several parameters associated with the process of photosynthetic induction were extracted from traces for individual

leaves, while other parameters were calculated. Each parameter was examined with ANOVA using a linear mixed-effects model with the fixed effect species and random effect study site, as described above. Extracted parameters included time to reach 90% full induction for A (t_{90-A}) and g_s (t_{90-g}), as well as the ratio of A to g_s under low light and full induction. Calculated parameters included those described above: one-point V_{cmax} , as well as the proportion of each photosynthetic induction trace subject to stomatal vs. biochemical limitations following leaf exposure to saturating light. As above, transformations indicated by the Box-Cox method (Box and Cox 1964) were required to meet assumptions of normality and equal variance for t_{90-A} ($1/y$), t_{90-g} ($y^{-0.5}$), V_{cmax} ($y^{-1.5}$), and the proportion of induction subject to stomatal limitations ($\sqrt{y}$).

Data were extracted from each leaf's lightfleck response to facilitate analyses. For both sets of lightfleck experiments, mean A and g_s from the last 30 s of each low light period were calculated, and maximum A, with its corresponding g_s , were determined for each saturating lightfleck. This approach was taken, in part, to accommodate system lags in the LI-6800. A repeated measures ANOVA was carried out for the saturating light portions of each lightfleck experiment where both A and g_s were examined in separate analyses as a function of species, lightfleck number (i.e., 1–5), and their interaction, with the leaf of each tree measured serving as the repeated measure. An identical analysis was carried out at low light for both lightfleck experiments, though there were 6 low light periods to analyze. Violations of sphericity were addressed by applying the Greenhouse–Geisser correction as needed (Greenhouse and Geisser 1959). Rates of A or g_s during the final lightfleck of each experiment (30 vs. 60 s) were compared via two-sample t-tests.

The predictors of A during leaf lightfleck experiments were examined using stepwise multiple regression analyses with backward and forward selection. Data from the first and fifth saturating lightflecks of both lightfleck experiments were analyzed to assess whether predictors of A varied temporally. Potential predictor variables included g_s during the prior low light period, g_s during the lightfleck's maximum rate of A, species, N_{area} (which is documented to affect photosynthetic capacity (Reich et al. 1998)), and LMA. Four separate multiple regressions were performed for each saturating lightfleck to sequentially exclude correlated predictor variables (i.e., only one g_s term was included at a time and N_{area} or LMA were also individually included). None of the predictor variables appearing in final models were significantly correlated, and all had Pearson correlation coefficients ≤ 0.27 (data not shown). The final selected multiple regression model for each analysis was that which minimized the Akaike Information Criterion (AIC). Partial correlations for predictor variables appearing in final models were calculated.

Induction state percentage (IS%) was calculated for both the photosynthetic and stomatal conductance rates of each species during each applied lightfleck as,

$$IS\% = \frac{(\text{rate}_{LF} - \text{rate}_{initial})}{(\text{rate}_{max} - \text{rate}_{initial})} 100\% \quad (2)$$

where rate_{LF} is maximum A or g_s achieved during a given lightfleck, $\text{rate}_{initial}$ is A or g_s during the 30 s of low light prior to any lightfleck application, and rate_{max} is A or g_s at full induction, under saturating light (Chazdon & Pearcy 1986). A single mean A_{max} or g_{s-max} value calculated from induction measurements was used for each species. Statistical analyses were not performed for IS% values, as several input terms were previously analyzed and/or were identical within a species or individual, thereby depressing variance.

Cumulative carbon assimilation (CCA), after Sakoda et al. (2021), was computed for each leaf measured in the lightfleck experiments while exposed to saturating light (i.e., during the five lightflecks, see Online Resource 2 for more detail). Also determined was CCA during photosynthetic induction (for the first 150 s or 300 s) when leaves were exposed to constant saturating light. To examine whether the rate of CCA varied for the two lightfleck experiments (five 30 s or 60 s lightflecks) when compared to the induction curves (150 s or 300 s of data), individual two-tailed t-tests were carried out by species for each lightfleck experiment.

Results

Understory light environment

Similar understory light environments were observed across trees and study sites. Specifically, GSF and percent canopy cover did not vary significantly ($p > 0.05$) with study site, tree species, or their interaction (Online Resource 3). The GSF term was 0.12 ± 0.006 (mean $\pm$ SE) across species and sites, while percent canopy cover was $91.1\% \pm 0.2$.

Leaf traits

Significant differences among species were observed for both LMA ($p < 0.0001$, Fig. 1a) and N_{area} ($p < 0.05$, Fig. 1b). Multiple comparisons indicated that LMA was highest for *A. rubrum*, followed by *A. platanooides*, then *F. grandifolia* ($p < 0.01$, Fig. 1a). Nitrogen per unit leaf area was also highest for *A. rubrum*, though only *A. rubrum* and *A. platanooides* differed significantly ($p < 0.05$, Fig. 1b). *Fagus grandifolia* had N_{area} values that were intermediate to and not significantly different from either of the *Acer* species ($p \geq 0.05$, Fig. 1b).

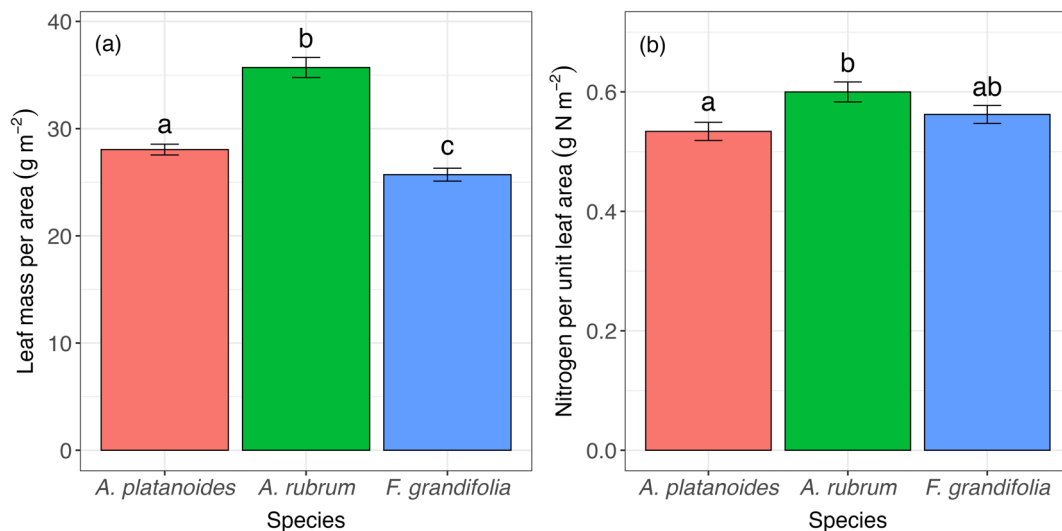


Fig. 1 Mean $\pm$ one SE of leaf mass per area (**a**, g m⁻²) and nitrogen per unit leaf area (**b**, g N m⁻²) for each study species. Species-level data were compiled across all experiments. Significant differences among

species were observed for both variables, and different lower-case letters indicate significant differences among species

Photosynthetic induction

Logistic regression model fits and comparisons showed that the photosynthetic induction curve of *F. grandifolia* differed significantly from an overall model fit to the induction responses of all three species, due in part to lower variance in the model fit (Fig. 2, Table 1). *Acer platanoides* achieved the highest modeled A under full induction (Y_{asym} , 4.32 $\mu\text{mol m}^{-2} \text{s}^{-1}$), followed by *F. grandifolia* and *A. rubrum* (3.52 and 3.48 $\mu\text{mol m}^{-2} \text{s}^{-1}$, respectively, Fig. 2, Table 1). The rate of increase in A after application of saturating light (k) was highest for *F. grandifolia*, and the species also displayed the shortest time required to achieve the maximum rate of increase in A (t_m , 314 s, Table 1). *Acer platanoides* had the longest time required to maximize the rate of increase in A (436 s), with *A. rubrum* intermediate (400 s, Table 1). Evident in raw leaf traces is high within-species variation, as well as substantial within-site variation in traces for all three species (Fig. 2a–c).

Despite species differences in the logistic regression analysis of A vs. time, the measured (rather than modeled) time to reach 90% of full induction (t_{90}) for A and g_s was invariant among species ($p > 0.05$), though variation in t_{90-g} was worth noting ($p = 0.0802$, Table 2). *Acer rubrum* had a t_{90-g} that was nearly half that (i.e., it reached 90% of maximum g_s more quickly) of the other species. Calculated one-point V_{cmax} was invariant among species ($p > 0.05$, Table 2). When assessing the proportion of photosynthetic induction curves subject to stomatal vs. biochemical limitations, significant differences among species were detected ($p < 0.001$), and it was *A. rubrum* that

differed from the other species (Table 2, Online Resource 4). The species had a significantly lower proportion of stomatal limitation (26%) and a correspondingly higher biochemical limitation (74%), relative to *A. platanoides* and *F. grandifolia*. The latter two species had a somewhat more even balance between stomatal and biochemical limitations during photosynthetic induction, at 54 and 46% for *A. platanoides* and 62 and 38% for *F. grandifolia*, respectively (Table 2, Online Resource 4).

To assess each species' water-use strategy during induction, A/g_s was computed at two timepoints; during the final 2 min of low light (10 $\mu\text{mol m}^{-2} \text{s}^{-1}$) and at full induction under saturating light (1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$), once g_s had stabilized. All three species maintained similar ratios under low light ($p > 0.05$), but significant differences among species emerged under saturating light ($p = 0.011$, Table 2). Again, it was *A. rubrum* that differed from the other species, with a significantly higher A/g_s (Table 2). A lower g_s was largely responsible for this difference (Table 2). In examining these values, it was evident that mean values of A at full induction were slightly lower (0.19–0.29 $\mu\text{mol m}^{-2} \text{s}^{-1}$ difference) than values modeled with logistic regression (Tables 1, 2). *Acer platanoides* and *A. rubrum* had the highest and lowest measured mean A values, respectively, with *F. grandifolia* intermediate (Table 2). This pattern is consistent with that found in the logistic regression analysis, and the discrepancy in measured vs. modeled A at full induction likely arose because there were fewer data points (i.e., fewer temporally long traces) as more time progressed on the logistic regression x-axis (Fig. 2, Table 1).

Fig. 2 Photosynthetic induction responses modeled with logistic regression for each species individually and all species pooled. Individual traces for each leaf measured are shown for *A. platanoides* (a), *A. rubrum* (b), and *F. grandifolia* (c), as well as best fit logistic regression lines and shaded 95% confidence intervals. Variation in leaf trace shading indicates leaves measured at different study sites. Whole-model F-tests indicated significant differences between the ‘All species’ model (black line and grey shading, d) and the *F. grandifolia* model (blue line and shading), while no difference was observed for either *A. platanoides* (red line and shading) or *A. rubrum* (green line and shading)

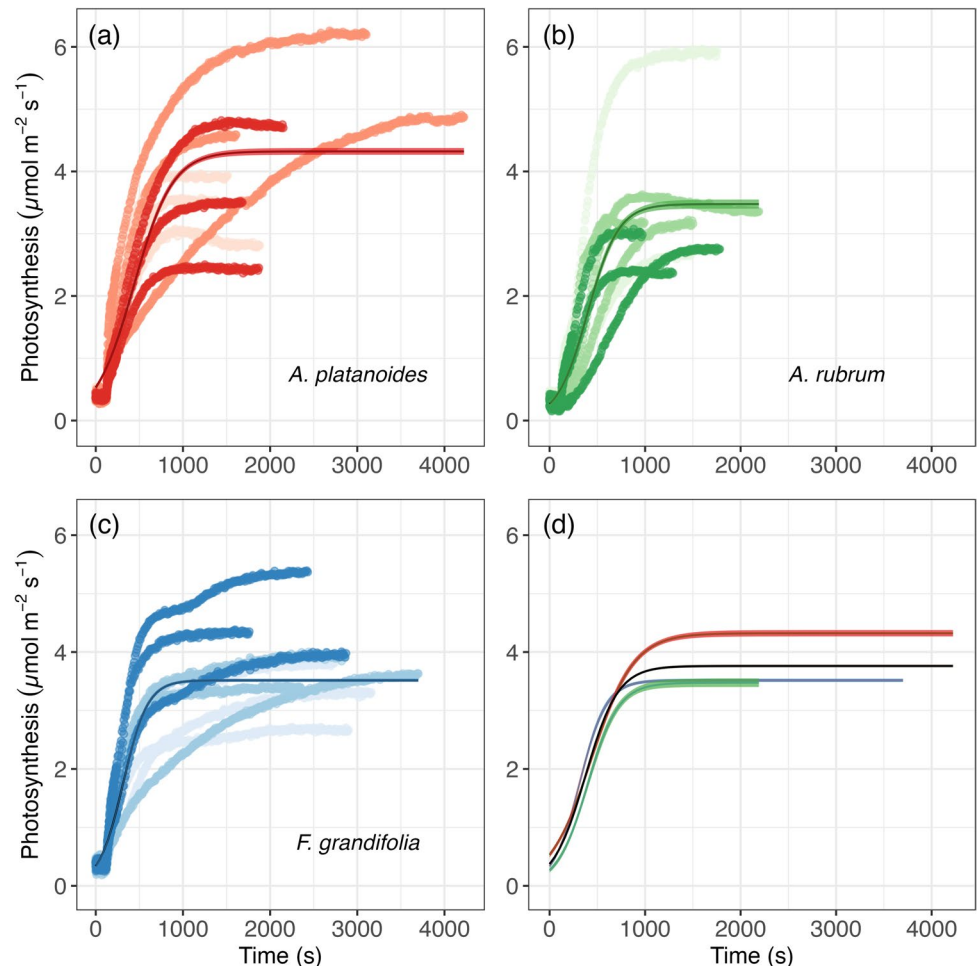


Table 1 Parameter estimates with 95% confidence intervals reported in parentheses and statistics for logistic regression analyses used to characterize induction responses of each species, as well as parameter estimates for a full model including all three species

Parameter	<i>A. platanoides</i>	<i>A. rubrum</i>	<i>F. grandifolia</i>	Full model
Y_{asym}	4.322	3.476	3.516	3.762
95% CI	(4.257, 4.389)	(3.400, 3.554)	(3.487, 3.545)	(3.733, 3.791)
k	0.00452	0.00623	0.00715	0.00588
95% CI	(0.00418, 0.00489)	(0.00573, 0.00679)	(0.00673, 0.00761)	(0.00564, 0.00613)
t_m	435.7	399.7	314.3	375.4
95% CI	(415.5, 457.2)	(381.0, 419.4)	(304.5, 324.4)	(366.5, 384.5)
RSE	0.8093	0.7899	0.6063	0.7674
df	2809	1976	3344	8135
p -value	0.9833	1	<0.0001	NA

Species-level models were compared to the full model with whole-model F-tests, with results reported in the final row

Lightfleck experiments—responses of A and g_s

The two lightfleck experiments, one with five sequential 30 s lightflecks applied, and the other with 60 s lightflecks, showed similar overall patterns in the responses of A and g_s during the saturating ($1000 \mu\text{mol m}^{-2} \text{s}^{-1}$) and low ($10 \mu\text{mol m}^{-2} \text{s}^{-1}$) light portions of the experiments (see

Online Resources 5 and 6 to view raw data traces for these experiments). Under saturating light in both experiments, A did not vary significantly by species ($p > 0.05$), though it increased significantly as subsequent lightflecks were applied ($p < 0.0001$, Fig. 3a, c, Table 3). A significant species by lightfleck interaction was found in both experiments ($p < 0.001$), indicating divergent species-level responses to

Table 2 Results of ANOVAs to assess variation in photosynthetic induction extracted or calculated parameters (mean $\pm$ one SE) for each tree species

Parameter	<i>A. platanoides</i>	<i>A. rubrum</i>	<i>F. grandifolia</i>	p-value
t_{90-A} (min)	10.84 ± 3.14	7.88 ± 1.29	10.95 ± 2.39	0.5113
t_{90-g} (min)	13.64 ± 4.32	6.80 ± 1.15	13.12 ± 2.60	0.0802
t_{90-V} (min)	7.89 ± 1.81	8.21 ± 1.21	8.01 ± 1.96	0.7275
Calculated one-point V_{cmax} ($\mu\text{mol m}^{-2} \text{s}^{-1}$)	19.48 ± 2.47	17.24 ± 1.97	19.81 ± 2.01	0.6544
Proportion stomatal limitation**	0.54 ± 0.08^a	0.26 ± 0.04^b	0.62 ± 0.02^a	0.0001
Proportion biochemical limitation**	0.46 ± 0.08^a	0.74 ± 0.04^b	0.38 ± 0.02^a	0.0003
A/g_s at low light ($\mu\text{mol mol}^{-1}$)	35.87 ± 2.98	33.07 ± 2.59	30.99 ± 2.23	0.4204
A/g_s at full induction ($\mu\text{mol mol}^{-1}$)*	68.20 ± 4.36^a	84.58 ± 3.11^b	70.30 ± 3.23^a	0.0110
A at low light ($\mu\text{mol m}^{-2} \text{s}^{-1}$)	0.40 ± 0.02	0.33 ± 0.02	0.39 ± 0.02	NA
g_s at low light ($\text{mol m}^{-2} \text{s}^{-1}$)	0.012 ± 0.0015	0.010 ± 0.0008	0.013 ± 0.0013	NA
A at full induction ($\mu\text{mol m}^{-2} \text{s}^{-1}$)	4.06 ± 0.39	3.29 ± 0.39	3.81 ± 0.25	NA
g_s at full induction ($\text{mol m}^{-2} \text{s}^{-1}$)	0.061 ± 0.0070	0.039 ± 0.0050	0.055 ± 0.0033	NA

Analyses assessed the time to 90% of the value at full induction for photosynthesis (t_{90-A}) and stomatal conductance to water vapor (t_{90-g}), as well as calculated one-point V_{cmax} , the calculated proportion of each induction curve exhibiting stomatal vs. biochemical limitation, and the ratios of photosynthesis to stomatal conductance to water vapor (A/g_s) at low light and full induction. Unanalyzed A and g_s at low light and full induction are also shown. Significant differences among species for a given parameter are indicated with asterisks (* ≤ 0.05 , ** ≤ 0.01 , *** ≤ 0.001) and superscripted letters

Fig. 3 Mean $\pm$ one SE photosynthesis ($\mu\text{mol m}^{-2} \text{s}^{-1}$) and stomatal conductance ($\text{mol m}^{-2} \text{s}^{-1}$) rates for each species observed during the 30 s (a, b) and 60 s (c, d) lightfleck experiments. Plotted points represent mean rates during the last 30 s of each low light period (PAR = $10 \mu\text{mol m}^{-2} \text{s}^{-1}$), occurring before and after lightflecks numbered 1–5 and the highest rate of photosynthesis and corresponding stomatal conductance achieved during each saturating light period (PAR = $1000 \mu\text{mol m}^{-2} \text{s}^{-1}$)

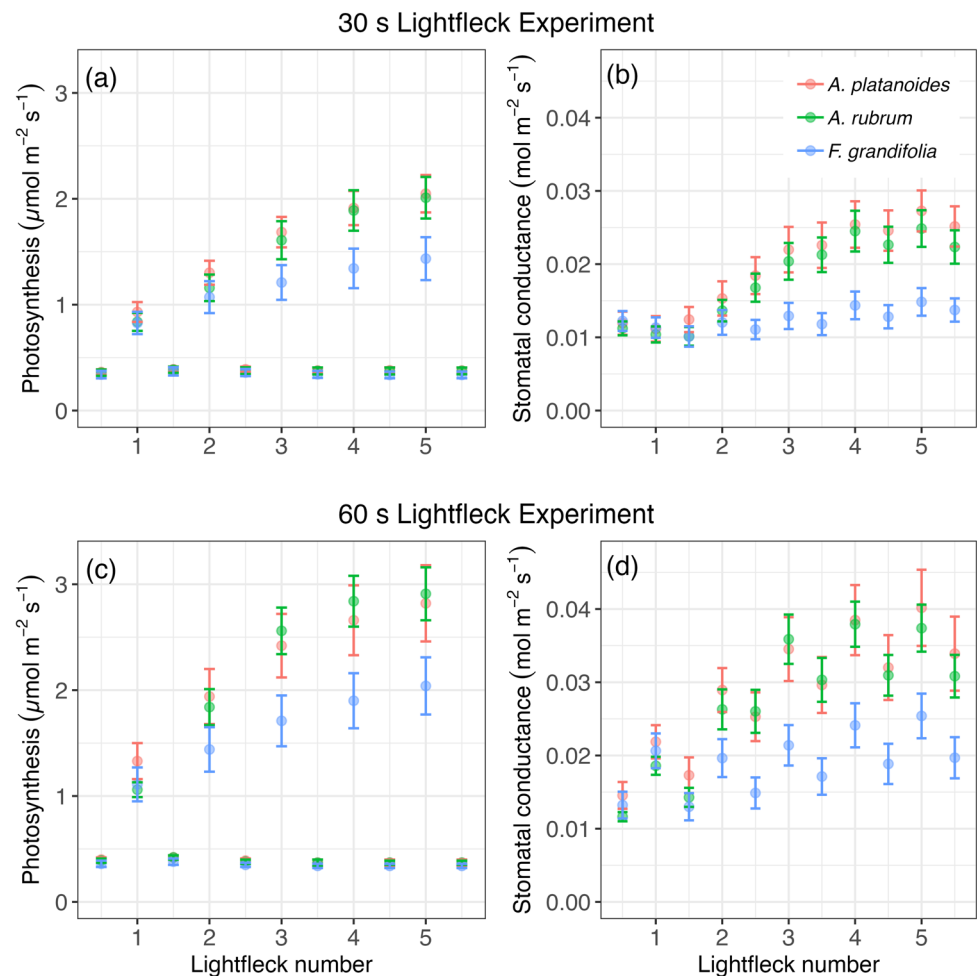


Table 3 Results of repeated measures ANOVA analyses to examine whether, during lightfleck experiments, photosynthetic rate (A) or stomatal conductance to water vapor (g_s) varied significantly by species, sequential lightfleck (lightflecks 1–5 at PAR = 1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$) or low light period (periods 1–6 at PAR = 10 $\mu\text{mol m}^{-2} \text{s}^{-1}$), or the interaction of these terms

	DF _n	DF _d	30 s lightfleck experiments			60 s lightfleck experiments		
			F	ε	<i>p</i> -value	F	ε	<i>p</i> -value
A at saturating light								
Species	2	24	1.99		0.1588	2.18		0.1354
Lightfleck	4	96	203.07	0.3108	<0.0001 [§]	182.58	0.3399	<0.0001 [§]
Species × Lightfleck	8	96	8.58		<0.001 [§]	7.98		<0.001 [§]
g _s at saturating light								
Species	2	24	3.21		0.0580	3.43		0.0490
Lightfleck	4	96	99.72	0.4735	<0.0001 [§]	65.15	0.5172	<0.0001 [§]
Species × Lightfleck	8	96	12.54		<0.0001 [§]	8.49		<0.0001 [§]
A at low light								
Species	2	24	0.33		0.7256	0.77		0.4726
Lightfleck	5	120	4.69	0.2795	0.0264 [§]	18.82	0.3138	<0.0001 [§]
Species × Lightfleck	10	120	0.57		0.6273 [§]	0.24		0.8768
g _s at low light								
Species	2	24	3.90		0.0341	3.56		0.0443
Lightfleck	5	120	108.79	0.3055	<0.0001 [§]	86.14	0.3595	<0.0001 [§]
Species × Lightfleck	10	120	17.64		<0.0001 [§]	8.18		<0.0001 [§]

Separate analyses were conducted for the 30 and 60 s lightfleck experiments, as well as for A and g_s during saturating lightflecks and low light periods. A [§] symbol indicates *p*-values upon which Greenhouse–Geisser corrections (ϵ) were applied due to unmet sphericity assumptions

applied lightflecks (Table 3). The two *Acer* species shared a similar response, with increased A as subsequent lightflecks were applied (Fig. 3a, c). However, *F. grandifolia* lagged, not displaying as dramatic an increase in A from lightflecks 1 through 5 (Fig. 3a, c). Under low light conditions preceding and following saturating lightflecks, a small but significant change in A was observed in both experiments from low light periods 1 through 6 ($p < 0.05$, Fig. 3a, c, Table 3). A slight increase in A (average increase of 0.025 $\mu\text{mol m}^{-2} \text{s}^{-1}$) from low light period 1–2 was observed in both experiments, before A returned to its approximate initial value in the 30 s experiment (average decrease of 0.009 $\mu\text{mol m}^{-2} \text{s}^{-1}$), or slightly lower than initial A in the 60 s experiment (average decrease of 0.025 $\mu\text{mol m}^{-2} \text{s}^{-1}$) by low light period 6. No species differences in A under low light were observed, nor were there significant interaction terms ($p > 0.05$, Table 3).

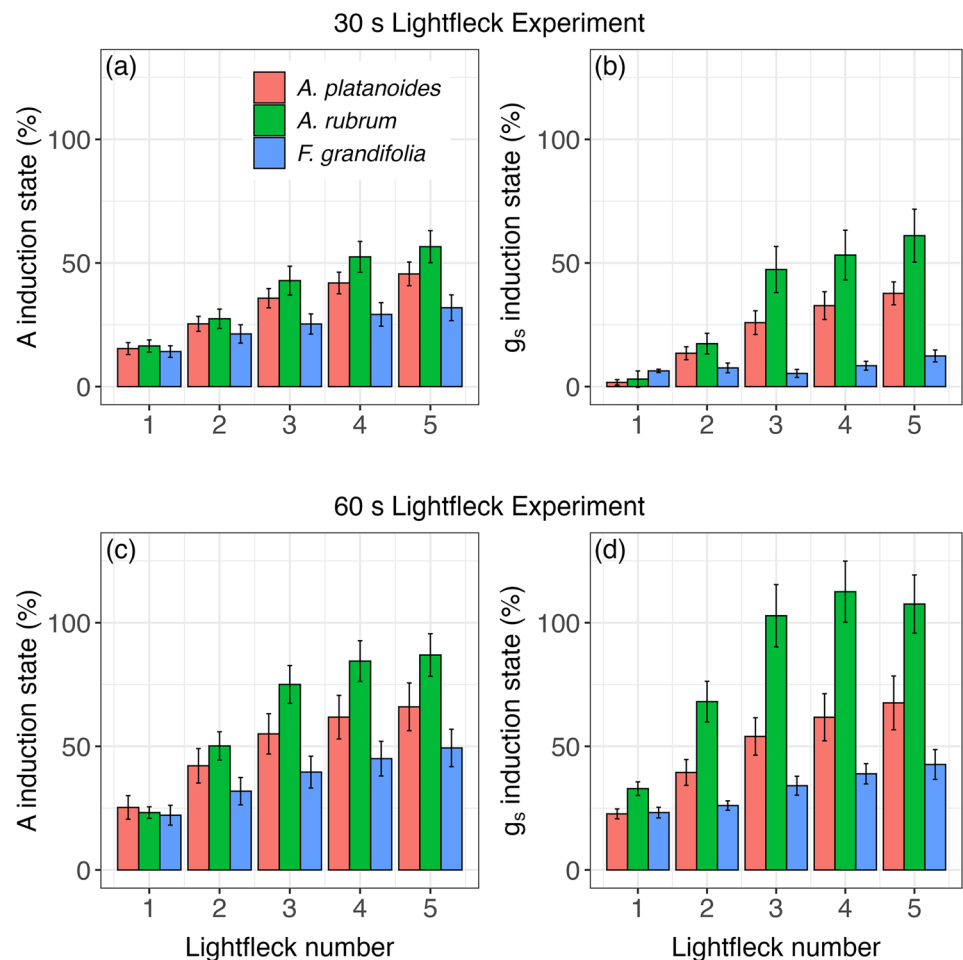
Some of the patterns in A were reinforced when observing stomatal responses during both lightfleck experiments. In particular, at saturating light g_s increased significantly with each lightfleck applied, and there was a significant interaction term for these analyses, with a *F. grandifolia* again displaying a relatively muted response to sequential lightfleck applications ($p < 0.0001$, Fig. 3b, d, Table 3). However, in contrast to the findings for A, notable species-specific responses in g_s were observed. Under saturating light, these responses were slightly more pronounced in the 60 s lightfleck experiment ($p = 0.0490$), relative to the 30 s experiment ($p = 0.0580$, Fig. 3b, d, Table 3), and both indicated higher g_s for the *Acer* species, when compared to

F. grandifolia during lightfleck exposure. An identical pattern was observed under exposure to low light conditions between lightflecks, where significant species, low light period, and interactive effects were all observed ($p < 0.05$). Again, it was *F. grandifolia* that lagged the *Acer* species as they responded with more dramatically increased g_s at each measurement interval (Fig. 3b, d, Table 3). Despite these differences, all three species demonstrated increased g_s at low light as the experiments proceeded from low light period 1 through 6 (Fig. 3b, d, Table 3).

It is evident that the 60 s lightfleck experiment led to higher rates of A and g_s during exposure to lightflecks compared to the 30 s experiment (Fig. 3a, c). By the fifth lightfleck, both A and g_s were significantly greater ($p < 0.01$) in the 60 s, relative to the 30 s experiment, with increases of 41% for A and 54% for g_s when data were pooled across species. Additionally, the amplitude of stomatal response to each subsequent lightfleck—low light period was higher in the 60 s, relative to the 30 s lightfleck experiment (Fig. 3b, d).

When A was examined as a function of A_{max} achieved during the induction experiments (i.e., in terms of IS%), it was again evident that longer duration lightflecks led to a more robust photosynthetic response (Fig. 4a, c). Overall, the patterns of change in IS% during saturating lightflecks reflected those described above, in which *F. grandifolia* was limited in its ability to increase A, relative to the *Acer* species (Fig. 4a, c). The highest IS% A values for all species were achieved during the fifth lightfleck of the

Fig. 4 Mean $\pm$ one SE induction state percentage (IS%) calculated for the 30 s (a, b) and 60 s (c, d) lightfleck experiments. The IS% was determined both for the maximum photosynthetic rate (A, a, c) and maximum stomatal conductance rate (g_s , b, d) achieved by each species during each lightfleck. The IS% term was calculated as the difference between the rate of A or g_s during a given lightfleck and its initial rate, prior to lightfleck application. This term was divided by the difference between A or g_s at full induction and the rate of A or g_s prior to lightfleck application. The quotient was multiplied by 100% to determine IS% during each applied lightfleck



60 s experiment. At this point, *A. rubrum* averaged 87% full induction, while *A. platanoides* averaged 66% and *F. grandifolia* reached an IS% of 49% (Fig. 4c). This species order, from highest to lowest IS% was maintained for nearly all saturating lightflecks across both experiments, with the only exception occurring during the first lightfleck of the 60 s experiment (Fig. 4a, c). Patterns of IS% for g_s largely mirrored those for A, in that the two *Acer* species increased IS% for g_s more rapidly than *F. grandifolia* in both lightfleck experiments (Fig. 4b, d). However, it is also evident that *A. rubrum*'s strong IS% response for A was supported by high g_s , relative to the other two species, exceeding an IS of 100% in the third, fourth, and fifth lightflecks of the 60 s experiment (Fig. 4).

Lightfleck experiments—predictors of A

The predictors of A during saturating light periods varied slightly across the two lightfleck experiments. For the 30 s lightflecks, both g_s during the immediately preceding low light period and N_{area} were the best predictors of A during the first and fifth periods of applied saturating

light ($p < 0.0001$, Table 4). Both variables were positively related to A (Table 4). Interestingly, g_s during the prior low light period explained more variation in A than g_s during a given lightfleck for both saturating light periods examined in the 30 s experiment (Table 4), despite the fact that these terms were positively correlated (Pearson correlation coefficient = 0.92, $p < 0.05$, data not shown). While the drivers of A did not change temporally in this experiment, these variables (i.e., g_s during the immediately preceding low light period and N_{area}) explained 51% of the variation in A during lightfleck 1 ($R^2_{adj} = 0.5077$) and increased to 62% for lightfleck 5 ($R^2_{adj} = 0.6154$, Table 4). Partial correlation analyses for these predictor variables showed that g_s consistently explained greater variance in A (partial correlation coefficients of 0.72 and 0.75 for lightflecks 1 and 5, respectively, Online Resource 7), relative to N_{area} (partial correlation coefficients of 0.30 and 0.46 for lightflecks 1 and 5, respectively, Online Resource 7).

During the 60 s lightfleck experiments, there was a temporal difference in the predictors of A from lightfleck 1–5. Eighty-one percent of the variation in A during lightfleck 1 was explained by g_s during the prior low light period and

Table 4 Parameter estimates and statistics associated with multiple regression analyses conducted for the maximum photosynthetic rate achieved during the first and fifth (final) lightflecks of the 30 s and 60 s lightfleck experiments

Parameters & statistics	30 s Lightflecks		60 s Lightflecks	
	Lightfleck 1	Lightfleck 5	Lightfleck 1	Lightfleck 5
Intercept	− 0.0838	− 0.3802	− 0.6393	− 1.2350
g_s – prior low light period	0.0502***	0.0503***	0.0804***	NA
g_s – during lightfleck	NA	NA	NA	0.0600***
Species	NA	NA	NA	NA
N_{area}	0.6155 ^a	2.0770*	1.3667*	3.2182**
LMA	NA	NA	NA	NA
RSE	0.1948	0.3866	0.1869	0.3495
R^2_{adj}	0.5077	0.6154	0.8084	0.8623
F	14.4	21.8	55.84	82.42
p -value	<0.0001	<0.0001	<0.0001	<0.0001
AIC	− 85.52	− 48.50	− 87.76	− 53.95

The p -values of statistically significant parameters are indicated with asterisks (* ≤ 0.05 , ** ≤ 0.01 , *** ≤ 0.001), and parameters denoted by NA did not appear in the final model

^a N_{area} remained in the final model due to AIC-driven model selection, but the p -value for this model coefficient was not significant ($p=0.1421$)

N_{area} , as was the case above ($R^2_{adj}=0.8084$, $p<0.0001$, Table 4). However, by lightfleck 5, it was g_s during the lightfleck, as well as N_{area} that were the best predictor variables, and together they explained 86% of the variation in A ($R^2_{adj}=0.8623$, $p<0.0001$, Table 4). Again, the variables g_s and N_{area} were positively related to A (Table 4) and partial correlation analyses showed that g_s explained greater variation in A (partial correlation coefficients of 0.90 and 0.91 for lightflecks 1 and 5, respectively, Online Resource 7), relative to N_{area} (partial correlation coefficients of 0.49 and 0.56 for lightflecks 1 and 5, respectively, Online Resource 7).

In all four analyses presented here, species identity was consistently unimportant in predicting A during lightflecks (Table 4). Additionally, when considering leaf traits, N_{area}

was always a better predictor of variation in A during lightflecks than LMA (Table 4).

Cumulative carbon assimilation

Comparisons of CCA during the lightfleck experiments and comparable time periods from photosynthetic induction curves demonstrated that the two *Acer* species were often able to achieve higher rates of assimilation when exposed to variable rather than constant saturating light. In the 30 s lightfleck experiments, *A. platanoides* significantly increased CCA by 49.6% under variable, relative to constant saturating light ($p<0.01$), though this difference was not maintained in the 60 s experiment ($p=0.0995$, Table 5).

Table 5 Mean $\pm$ one SE cumulative carbon assimilation (CCA, $\mu\text{mol m}^{-2}$) during induction and lightfleck experiments

Species	CCA—induction ($\mu\text{mol m}^{-2} \text{ s}^{-1}$)	CCA—lightfleck ($\mu\text{mol m}^{-2} \text{ s}^{-1}$)	Mean CCA difference (lightfleck – induction)	Increase in CCA relative to induction	p -value
30 s lightfleck comparison					
<i>Acer platanoides</i> **	179.3 $\pm$ 18.9	268.2 $\pm$ 21.2	88.9	49.6%	0.0065
<i>Acer rubrum</i> ***	116.4 $\pm$ 17.7	256.9 $\pm$ 23.9	140.5	120.7%	0.0003
<i>Fagus grandifolia</i>	172.9 $\pm$ 12.4	206.6 $\pm$ 27.5	33.7	19.5%	0.2926
60 s lightfleck comparison					
<i>Acer platanoides</i>	491.7 $\pm$ 48.5	654.3 $\pm$ 77.9	162.6	33.1%	0.0995
<i>Acer rubrum</i> ***	350.5 $\pm$ 60.4	656.5 $\pm$ 50.4	306.1	87.3%	0.0016
<i>Fagus grandifolia</i>	479.5 $\pm$ 43.7	482.5 $\pm$ 65.7	3.0	0.6%	0.9711

The CCA during induction was determined for the 150 s following exposure to saturating light (CCA-induction) for comparison with the 30 s lightfleck experiments (150 s total of saturating light applied, CCA-lightfleck), while CCA during induction was calculated for the first 300 s following exposure to saturating light for comparison with the 60 s lightfleck experiments (300 s total of saturating light applied). Within a species, t -tests were used to assess differences in CCA-induction vs. CCA-lightfleck for both the 30 and 60 s lightfleck experiments. Significant differences are indicated with asterisks (* ≤ 0.05 , ** ≤ 0.01 , *** ≤ 0.001). Also reported are the mean difference between CCA-lightfleck and CCA-induction, as well as the percent increase in CCA-lightfleck, relative to CCA-induction

Acer rubrum exhibited significant increases in CCA in both the 30 and 60 s lightfleck experiments, with increases of 120.7 and 87.3%, respectively ($p < 0.01$, Table 5). The CCA of *F. grandifolia* did not differ significantly between variable and constant saturating light conditions ($p > 0.05$, Table 5).

Discussion

Photosynthetic induction

Contrary to expectations, photosynthetic induction responses were similar among study species, with measured and modeled $A_{\max}$ for fully induced leaves varying by $< 0.9 \mu\text{mol m}^{-2} \text{s}^{-1}$ (Tables 1, 2, Fig. 2). Although, *F. grandifolia* exhibited an earlier (t_m) and faster (k) modeled response to saturating light, relative to the *Acer* species, this difference was attenuated by the time 90% full induction was reached (Tables 1, 2). Given the shared low-light environment occupied by these species and relative similarities in shade tolerance, such findings are aligned with long-held assumptions about shared induction response patterns among species from similar light environments (Naumburg and Ellsworth 2000; Vico et al. 2011). Despite these similarities, the physiological strategies employed by the three tree species diverged, with *A. rubrum* displaying different limitation patterns during induction, as well as more conservative water use.

Proportional biochemical limitations during photosynthetic induction were significantly greater for *A. rubrum* (74%) than for *A. platanoides* (46%) and *F. grandifolia* (38%), while the opposite was true for stomatal limitations (Table 2). At full induction, g_s was 29–36% lower for *A. rubrum*, relative to the other study species and, though not significant ($p = 0.0802$), the time to 90% g_s was nearly half that of the other species. This indicates slightly more responsive stomata to applied saturating light in *A. rubrum*, but a more conservative g_s overall. Variation in leaf traits suggested that *A. rubrum* was likely to have the highest photosynthetic capacity of the three study species, given its high LMA and N_{area} (Fig. 1; Reich et al. 1998; Poorter et al. 2009) and assuming the study species partitioned leaf mass similarly (Osnas et al. 2018). Surprisingly, calculated one-point V_{cmax} , a measure of photosynthetic capacity, did not differ among species, though this may be due to high within-species variation in leaf traces, as well as error introduced by the one-point V_{cmax} method (Table 2, Fig. 2, De Kauwe et al. 2016). The strong biochemical limits on induction exhibited by *A. rubrum* may be due to the species' inability to rapidly activate Rubisco under conditions of low intercellular CO_2 concentration (c_i), given moderate g_s (Way and Pearcy 2012). Counterbalancing high biochemical limits, *A. rubrum*'s significantly lower stomatal limitations on

induction were likely supported by the species' rapid rate of stomatal opening. Once induced, a higher photosynthetic capacity could support the species' significantly higher A/g_s , thus allowing for greater water conservation relative to the other study species.

In contrast, *A. platanoides* and *F. grandifolia*, exhibited significantly lower A/g_s under full induction, driven by higher g_s , and experienced significantly greater proportional stomatal limitation during induction (Table 2). It has been suggested that such a gas exchange strategy may serve as a "catch-up measure" to offset stomatally-limited carbon gain during induction or simply to maximize A by maintaining higher c_i under saturating light (Deans et al. 2019). Either of these possibilities, both supporting higher $A_{\max}$ via enhanced g_s , would help to offset lower photosynthetic capacity in these species, relative to *A. rubrum* (evidenced by significantly lower LMA in both species, as well as significantly lower N_{area} for *A. platanoides*, relative to *A. rubrum*, Fig. 1). While g_s is a key factor controlling leaf induction responses after biochemical limitations have been met (Way and Pearcy 2012), it also likely plays a role in modulating the effect of differences in leaf traits for these understory saplings. Despite physiological differences among the study species, similar overall induction responses provided no evidence that *A. platanoides* held a particular advantage over the two native tree saplings.

Several studies, spanning variable climates and plant functional types, report that g_s prior to induction exerts substantial control over induction rate in shade-tolerant species (Allen and Pearcy 2000; Santiago and Dawson 2014; Wachendorf and Küppers 2017). Both Allen and Pearcy (2000) and Wachendorf and Küppers (2017) identified threshold initial g_s values (~ 0.010 – $0.030 \text{ mol m}^{-2} \text{s}^{-1}$), above which induction proceeded much more quickly. Although g_s values measured in low light were similarly low among species in the present study (0.010 – $0.013 \text{ mol m}^{-2} \text{s}^{-1}$), time to 90% induction was relatively rapid (≤ 14 min for all species) compared to prior reports (Table 2; Santiago and Dawson 2014; Wachendorf and Küppers 2017). Naumburg and Ellsworth (2000) also observed *A. rubrum* to demonstrate this anomalous pattern.

Rapid induction responses following exposure to high light is in line with theoretical ideas regarding shade-tolerant plants (e.g. Vico et al. 2011; Way and Pearcy 2012). Specifically, the leaves of such species are expected to readily respond to intermittent light availability, without necessarily depending on high g_s in low light to facilitate such a response. *Acer rubrum*, *A. platanoides*, and *F. grandifolia* all exhibited these relatively rapid responses to saturating light exposure, but identifying overarching patterns associated with shade tolerance has proven difficult. Vico et al.'s (2011) meta-analysis of stomatal delays associated with induction failed to find any relation

between variation in shade tolerance and stomatal responsiveness to variable light. However, the authors noted that methodological differences among studies may confound the ability to draw such conclusions. Others have suggested that an examination of species-specific differences in water-use efficiency may be central in developing a more unified understanding of variation in g_s during photosynthetic induction (Knapp and Smith 1989; Naumburg and Ellsworth 2000).

Photosynthetic induction versus lightfleck responses

The modeled rate of induction gain varied only moderately by species, as did measured and modeled values of $A_{\max}$ (Tables 1, 2, Fig. 2). These similarities were not apparent in the lightfleck experiments, during which both *Acer* species were more responsive, in terms of A and g_s , as well as IS%, to short duration lightflecks (Table 3, Figs. 3, 4). The *Acer* species gained induction more quickly when exposed to lightflecks, as evidenced by higher CCA, relative to conditions of constant saturating light (Table 5). Such stomatal responsiveness in support of A in fluctuating light environments is consistent with observations of *A. rubrum* as a species with highly responsive stomata, in which rapid induction gain and loss have been observed, as well as rapid responses to lightflecks by previously induced leaves (Naumburg and Ellsworth 2000). It may be that some congeners, such as *A. platanoides*, share these traits, though the pattern is not universal within the genus, as variable results have been documented for different shade-grown *Acer* species (Lei and Lechowicz 1997; Urban et al. 2007). In contrast, *F. grandifolia*'s lightfleck response closely mirrored its induction response, demonstrating comparable CCA across experiments (Table 5).

Similarities among the species following constant saturating light exposure was not an indication of similar responses in scenarios that better resemble typical forest understory conditions. Instead, the data indicate that species with more responsive stomata (i.e., the *Acer* species), able to maintain higher g_s during brief low-light periods between applied lightflecks, are most capable of rapid induction gain under variable light exposure. While there have been calls in the literature to incorporate induction-related lags into what are typically steady-state models of photosynthesis (e.g. Porcar-Castell and Palmroth 2012; Way and Pearcy 2012), this comparison of induction curves and lightfleck responses indicates that reliance solely on induction curves and derived parameters such as time constants will not adequately capture leaf responses to the fluctuating light environment of forest understories.

Lightfleck experiments—responses of A and g_s

As observed for other shade-tolerant woody plants growing in forest understory environments, the IS% for A continually increased in leaves exposed to successive lightflecks, regardless of duration (Fig. 4; Chazdon and Pearcy 1986; Rijkers et al. 2000). This indicated that all study species were capable of gaining induction when exposed to naturally-occurring sunflecks, which are often temporally clustered and short in duration (Chazdon 1988; Pfitsch and Pearcy 1989a; Pearcy et al. 1994). However, as noted above, the *Acer* species gained induction more quickly during lightflecks than when exposed to constant saturating light. This resulted in significantly enhanced CCA, particularly for the shorter duration 30 s lightflecks, where *A. platanoides* and *A. rubrum* respectively demonstrated 50% and 121% greater carbon gain relative to conditions of constant saturating light (Table 5). High g_s during and between lightflecks supported rapid induction gain in both *Acer* species, relative to *F. grandifolia* (Table 3, Fig. 3).

Previously published work shows some evidence for this pattern of more rapid induction gain under dynamic light in tropical understory tree saplings (Rijkers et al. 2000), though not in a tropical shrub (Chazdon and Pearcy 1986). It has been argued that some species maintain high g_s in low light because water loss is outweighed by potential carbon gain via the maintenance of high c_i , as well as increased quantum yield upon light exposure (Pfitsch and Pearcy 1989b). This strategy appears to have been employed by the *Acer* species in response to sequential lightflecks, while *F. grandifolia* remained more conservative, with lower g_s , thus A , throughout both lightfleck experiments. This pattern was particularly noteworthy for *A. rubrum* given the species' relatively conservative water use (i.e., A/g_s) once fully induced. Exposure to lightflecks instead appears to have triggered *A. rubrum*'s responsive stomata, as g_s during 60 s lightflecks often exceeded g_s observed under full induction (Fig. 4).

Findings for the *Acer* species are consistent with recent research showing that plants with mutations or genetic modifications to enhance g_s and/or stomatal responsiveness also demonstrate fast induction gain (Kimura et al. 2020; Yamori et al. 2020). Shared stomatal characteristics may underlie the *Acer* species high stomatal responsiveness to lightflecks. In particular, small stomatal size, often coupled with high stomatal density, enhances stomatal responsiveness to light in species with similar stomatal anatomy, due in part to a shorter diffusion pathway and lower energetic demands associated with guard cell movements (Franks and Beerling 2009; Drake et al. 2013; McAusland et al. 2016; Lawson and Violet-Chabrand 2019; Yoshiyama et al. 2024).

Exposure to longer duration (60 s) lightflecks resulted in significantly greater induction gain for all study species by the fifth lightfleck, indicating more rapid attenuation of

biochemical and stomatal induction limitations than when 30 s lightflecks were applied. However, overall patterns of A and g_s during both lightfleck experiments showed consistent species-specific responses to lightfleck application. Despite achieving a higher IS% for A during lightfleck application, *A. rubrum* had similar CCA to *A. platanoides* in these experiments, likely due to *A. rubrum*'s slightly lower mean $A_{\max}$ ($0.77 \mu\text{mol m}^{-2} \text{s}^{-1}$ lower, Table 5, Fig. 4). These findings indicate both species are well positioned to thrive in the forest understory's variable light environment. One area in need of further study is the persistence of induction once leaves return to diffuse light conditions, as this will temper the ability of leaves to respond to natural sunflecks throughout the day.

Lightfleck experiments—predictors of A

Examination of the main predictors of A during the first and fifth lightflecks in these experiments showed that lightfleck duration affected stomatal responses (Table 4). Specifically, g_s during low light periods between lightflecks best predicted A in the 30 s experiment, as well as during the first lightfleck of the 60 s experiment. However, by the fifth lightfleck of the 60 s experiment, stomata were more responsive, and it was g_s during the fifth 60 s lightfleck that best predicted A (Table 4, Fig. 3). The progressive and significant rise in g_s demonstrates the importance of maintaining this aspect of induction during brief (2 min) low light periods, though it also shows that these shade-tolerant species vary in their adaptations to such a light environment. The *Acer* species were significantly more profligate in water use (assessed via g_s) than *F. grandifolia*, during lightfleck exposure (Table 3).

In addition to g_s , N_{area} was also an important predictor of A during the lightfleck experiments, while species identity did not appear in any of the models (Table 4). This indicates that leaf-level variation in g_s and N_{area} were more critical predictors of A than was species identity. Few studies of induction and lightfleck responses in woody plants have included chemical or morphological leaf trait data (though see Küppers et al. 1996; Santiago and Dawson 2014), and to our knowledge, none have sought to quantitatively predict drivers of A during lightfleck exposure. Recent literature highlights the importance of intraspecific trait variation (Violle et al. 2012; Siefert et al. 2015), as it can sometimes be of a magnitude that is comparable to interspecific variation (Messier et al. 2010). A combination of genotypic variation, phenotypic plasticity, and their interaction can drive high intraspecific trait variation (Albert et al. 2011), and leaf chemical traits, including N_{area} , have been identified as particularly variable within species (Rozendaal et al. 2006; Siefert et al. 2015).

While the absolute range of variation in N_{area} was relatively restricted among our study species, the multiple

regression analyses indicated that intraspecific variation was meaningfully large (Table 4, Fig. 1). Variation in traits including N_{area} and LMA are well documented within individual trees of a given species, but this variation is typically related to vertical gradients of light, temperature, vapor pressure deficit, and gravity (Niinemets 1997; Baldocchi et al. 2002; Duursma and Marshall 2006; Poorter et al. 2009). The present study points to the existence of an additional explanation; it may be that the heterogenous conditions caused by a dynamic light environment, together with genetic variation (three populations per species were sampled) and variable phenotypic plasticity, lend themselves to high intraspecific trait variation. These considerations may also underlie the intraspecific variation observed in overall induction responses (Fig. 2). If this is the case, it may help clarify the longstanding difficulty in drawing clear conclusions about the induction responses of shade-tolerant species as a whole. Given that 86% of the variation in A during the final lightfleck of the 60 s experiment was captured by just measures of g_s and N_{area} (Table 4), it is worth reconsidering how we think about organizing woody plants into categories as we attempt to characterize induction and lightfleck responses.

Conclusions

Photosynthetic induction responses of the three focal species showed surprisingly contrasting results, when compared to lightfleck responses. Interspecific differences in induction responses were limited, though largely uncoupled from lightfleck responses, where *F. grandifolia* was observed to lag the *Acer* species. Contrary to expectations, all three species gained induction at a similar rate and magnitude when exposed to constant saturating light; however, the two *Acer* species gained induction more rapidly, exhibiting higher CCA, when exposed to conditions intended to mimic forest understory light environments due to high stomatal responsiveness. In contrast, *F. grandifolia* gained induction at a similar rate, regardless of whether the species was exposed to lightflecks or constant saturating light. These contrasting lightfleck responses could have significant implications for total plant carbon gain in low-light environments, if they are found to be widespread.

Both *Acer* species are clearly well-adapted to efficiently use lightflecks in support of carbon gain, and nothing in the present study indicates that *A. platanoides*, despite its characterization as an aggressive competitor, holds a particular ecophysiological advantage over its congener. However, without intervention, *A. platanoides* is likely to maintain a presence in forest understories throughout North America, contributing to regeneration mismatch in many regions. While the limited ability of *F. grandifolia* to respond quickly to lightflecks may reduce overall carbon gain and growth

over time, the species is also under substantial threat from both beech bark disease and beech leaf disease in North America. These introduced diseases, the latter of which is not yet fully understood, are accelerating *F. grandifolia* mortality, and will ultimately increase habitat space in eastern broadleaf forests (Reed et al. 2022). Both *Acer* species appear well positioned to succeed *F. grandifolia*, as its likely decline proceeds.

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Author contributions JLS conceived the project, carried out field work, performed data analyses, and drafted the manuscript. SP conducted field work, laboratory sample processing and analysis, data processing, and contributed to manuscript development.

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Declarations

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